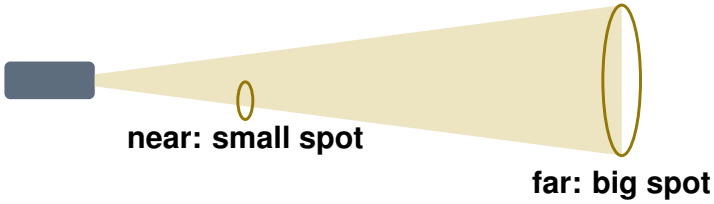
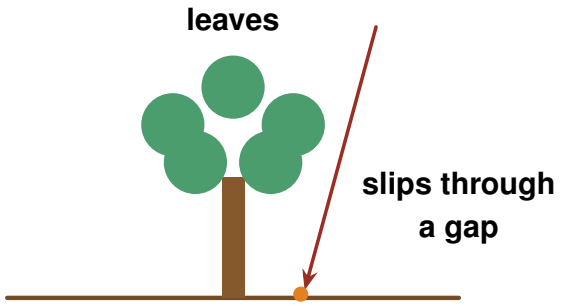
The Laser Beam and the Echo

BIG IDEA

A laser spot grows with distance, and LiDAR sees through gaps, never through solid things.





THE IDEA

Shine a flashlight at your hand and you see a small circle. Shine it at a far wall and the circle is much bigger. A laser beam spreads too, just very slowly. Far away, the spot can be as big as a coin, so it can touch the edge of a pole and the wall behind it at the same time. The farther away something is, the dimmer its echo.

REMEMBER

- ✓ Twice as far away means only about one quarter as bright.
- ✓ Shiny white things send back lots of light. Black or wet things send back very little.
- ✓ LiDAR reaches the ground under trees only through gaps between the leaves. Water stops it.

TRY THIS

In a dark room, shine a flashlight at your hand, then at a far wall. Compare the size and brightness of the spot.

The full module is in the Reference Book.